Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. You are working with a stereo camera system. The right camera is located at the coordinates (3, 2, 1) in the world coordinate system. The relative transformation matrix from the right camera to the left camera is:

1 / 1 point

1	-0.5	0	1
2	0	1	2
-0.25	0.75	0.5	3
0	0	0	1

Find the coordinates of the left camera's position.

- (3, 9, 4.25)
- (4,4,4)
- (9, 3, 1)
- (3, 7.5, 9)

Correct

$X_l = T^*X_r$. T describes the rotation and translation between the right and left cameras.

$$\begin{bmatrix} x_l \\ y_l \\ z_l \\ 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix}$$

2. What is the primary challenge in 3D reconstruction from uncalibrated stereo images when dealing with textureless or reflective surfaces?

1 / 1 point

- Occlusion
- Noise in the images
- Lack of feature points
- Geometric distortion

Correct

The primary challenge when dealing with textureless or reflective surfaces in 3D reconstruction from uncalibrated stereo images is the "Lack of feature points." Textureless or reflective surfaces may not have distinct features that can be easily matched between images, making it challenging to compute accurate depth information.